



FUTURE SCHOLARS ETHIOPIA

PYTHON BASICS FOR BEGINNERS

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PART TWO





What will we learn?

- Variables
- Data Types
- User Input
- Mini Project



What is a Variable?

- A variable is a name used to store a value.
e.g., `name = "John"`
`age = 20`
- '=' here doesn't mean "is equal to", it means "Put this value into this variable."



Naming Variables

- Possible examples of variable names:

- `First_name = "Abebe"`
- `student_age = 17`
- `_my_score = 95`

- Impossible variable names:

- `2name = "Abebe"`
- `student age = 17`





Rules while Naming Variables

- Start ONLY with an alphabet or underscore (_)
- Use only Alphabates, numbers and underscores (_)
- No spaces
- Are case-sensitive (e.g., `name` is not equal with `Name`)





What is a Data Type?

- A data type tells Python what kind of value something is.

Data Types

- String
- Integer
- Float
- Boolean





1. String

- Strings are text.
 - Strings are written inside single quotes (' ') or double quotes (" ").
 - Anything you want Python to treat as text can be stored as a string. (e.g., "123")
- e.g., "Abebe"
"Ethiopia"
"Hello!"
"Pyhton is easy!!"





2. Integer

- Integers are any numbers.
- Has no quotation marks
- **15** is treated as a number (integer), but **"15"** is treated as a text by python.

- e.g., **1**
135
0





3. Float

- Floats are numbers with a decimal point.
 - Must have a decimal point
- e.g., 5.0
3.14
7.1





4. Boolean

- A Boolean has only two possible values:
 - True
 - False
- Booleans represent true/false information.





Checking Data Types

- By using `type()`, Python can tell us whether something is a string, integer, float, or Boolean.

e.g., `height = 1.75`

```
print(type(height))
```



User Input

- If we want the user to provide information, we use `input()`.
 - e.g., `input("What is your name: ")`
`input("What subject do you like: ")`
- Assigning `input()` into a variable is useful so that you can do something with the input.
 - e.g., `name = input("What is your name: ")`
`print("Hello", name)`



- `input()` gives us a string.
- Even if the user enters an integer (e.g., **12**), python receives **"12"**.
 - e.g., `name = input("How old are you: ")`
`print = (type(age))`



Data Type Conversion

- We can use `str()`, `int()`, `float()`, `bool()` to change a data into string, integer, float, or boolean respectively.
 - e.g., `name = int(input("How old are you: "))`
`print = (type(age))`



Mini Project

- Create a Python program that asks a user for their
 - full name
 - age
 - school name
 - grade
 - height
- After collecting all the information, use `print()` to display the information in a clear and organized Personal Profile.

